Personalizing POI Descriptions with LLMs: A Study of User Perceptions and Informational Value

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Abstract

Many Points of Interest (POI) descriptions remain generic and static, presenting identical content to all users regardless of interests or travel preferences. This can limit perceived relevance and may reduce engagement with travel platforms. This study examines whether large language models (LLMs) can personalize POI descriptions to improve perceived information quality. In a within-subjects study, 51 participants evaluated original and LLM-adapted descriptions across 10 POIs. Pooled rating distributions show improved perceived clarity ($p = 0.0218$, uncorrected) and borderline evidence of improved trustworthiness ($p = 0.050$), while perceived significance remained comparable ($p = 0.30$). Optional post-study responses indicate relatively high comfort with AI-generated travel content (mean = 3.94, $n = 17$). These findings suggest that LLM-based personalization can improve perceived clarity in tourism micro-content without clear losses in perceived credibility.

CCS Concepts

• **Human-centered computing → User centered design**; *User studies.*

Keywords

poi adaptation, LLM-generated content, ai-driven descriptions, content personalization

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1 Introduction

Travel planning increasingly happens through digital platforms that help users evaluate accommodations, attractions, and other Points of Interest (POIs). Recommender systems have therefore become central in tourism applications, where personalization is used to cope with information overload and heterogeneous user needs [1, 7]. In the POI domain specifically, prior work has largely emphasized predicting which places users may prefer (e.g., next-POI or ranked POI recommendation) and how to incorporate richer signals such as context and heterogeneous data [5, 10]. Recent surveys further argue that large language models (LLMs) are becoming an enabling component across the recommendation pipeline, including user interaction and explanation generation [8, 9, 13].

However, while ranking and recommendation determine *which* POIs are shown, users still rely on POI *descriptions* to decide *why* a place is relevant and whether it is credible and worth visiting. In many systems, these descriptions remain static and generic, presenting the same framing to all readers despite differences in interests, travel styles, and constraints. This "one-size-fits-all" presentation can under-emphasize what matters most to a given user (e.g., historical context versus family suitability), potentially limiting perceived clarity and usefulness. Generative AI offers a way to adapt the *emphasis* of descriptions at scale, complementing POI recommendation and supporting more user-centered travel decision-making [4, 12].

At the same time, deploying LLM-generated travel content raises practical concerns around information quality and trust, especially in domains where credibility is critical. Tourism-focused GenAI research has begun to examine opportunities and risks (e.g., reliability, domain adaptation, and user acceptance), and approaches such as retrieval augmentation have been proposed to mitigate factual errors [3, 4, 6]. Motivated by this gap, this study examines whether LLM-based personalization of POI descriptions can improve perceived information quality without harming trust or engagement. A prototype was developed that presents users with original (manual) descriptions and LLM-adapted variants conditioned on stated preferences, and both versions were evaluated in a within-subjects study across multiple POIs. The contributions of this study are (1) an empirical comparison of manual versus LLM-adapted POI descriptions using significance, trustworthiness, and clarity as evaluation dimensions, and (2) evidence about user acceptance of adaptive description personalization in a tourism context.

2 Related work

Prior work on tourism and location-based personalization has largely focused on point-of-interest (POI) recommendation, where systems predict or rank places that match a user profile and context. Surveys of travel and POI recommenders highlight the importance of heterogeneous signals such as mobility patterns, time, geography, and user attributes, as well as evaluation challenges that arise from contextual variability and data sparsity [7, 10]. Recent work in the UMAP community also emphasizes that user preferences can differ before and after a visit, which complicates modeling and suggests that preference signals and content expectations are not static [2].

With the rise of large language models (LLMs), research has expanded to interactive and generative paradigms for personalization, including LLM-based pipelines for POI recommendation and user profile generation [5, 8, 9, 11]. Tourism research has also begun evaluating generative AI for content creation and trip planning [3, 4, 12]. However, these works often study destination promotion or end-to-end planning rather than the micro-content users read when deciding whether a POI is relevant. This study addresses this

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gap by adapting POI descriptions to user interest profiles while keeping core facts stable, measuring perceptions of significance, trustworthiness, and clarity.

3 Methodology

This section outlines the methodology employed to evaluate the effectiveness of LLM-generated personalized POI descriptions compared to original manual descriptions. The study design, description personalization process, evaluation interface, metrics, procedure, and participant recruitment are described below.

3.1 Study Design

A within-subjects experimental design was employed in which each participant evaluated both original and AI-personalized descriptions for multiple POIs. This design enabled direct comparison between description types while controlling for individual differences in preferences and expectations. The study presented participants with 10 POIs spanning five categories: hotels, restaurants, museums, gardens, and landmarks. Each category included two POIs to ensure diverse representation of travel-related content. POIs were selected to include both internationally recognized locations and lesser-known destinations, providing variety in familiarity levels.

3.2 Description Personalization

For each POI, two description versions were prepared. The original description was sourced from official tourism websites and travel platforms, representing standard, non-personalized content. The AI-adapted version was generated using GPT-4o[1]. The model received a system prompt instructing it to act as an expert travel writer, emphasizing unique aspects, cultural significance, and visitor experience while adapting content to the user's situation. The user prompt included ten personalization dimensions: age, gender, marital status, whether the user has children, interests, travel experience level, education, profession, hobbies, and preferred travel style. The original POI title and description were provided, and the model was instructed to maintain factual accuracy while emphasizing aspects most relevant to the user's profile. For example, a museum description for a participant interested in history would emphasize historical significance and artifacts, while the same museum described for a family traveler would highlight interactive exhibits and educational programs.

3.3 Evaluation Interface

Participants viewed descriptions through a web-based interface that displayed two versions side by side, labeled as Version A and Version B (see Figure 1). The interface did not indicate which version was AI-generated. Each POI was accompanied by a representative image to provide visual context. The presentation order of AI and manual descriptions was randomized across POIs to prevent order bias. Participants could not proceed to the next POI without completing evaluations for the current one, ensuring complete data collection.

[1]https://platform.openai.com/docs/models/gpt-4o

3.4 Evaluation Metrics

Participants evaluated each description on three core dimensions using Likert scales:

Significance: "Does the description effectively communicate the significance and offerings of the place?" rated on a 5-point scale from Strongly Disagree to Strongly Agree. This metric assessed whether the description conveyed the value and appeal of the POI.

Trustworthiness: "How trustworthy does this description appear to be?" rated on a 5-point scale from Not at all to Extremely. This metric measured perceived credibility and reliability of the information.

Clarity: "How clear and complete is the information provided?" rated on a 5-point scale from Very Unclear to Very Clear. This metric evaluated information quality and comprehensibility.

After rating both descriptions individually, participants completed comparative evaluations selecting which description was more engaging, more informative, more appealing, and more likely to encourage a visit. Participants also reported familiarity with each POI using three categories: visited in person, heard about it, or never heard of it.

3.5 Statistical Analysis

Results are summarized using pooled response distributions across all POIs and participants ($N = 510$ ratings per condition for each metric). Differences between manual and AI rating distributions were evaluated using chi-square tests of independence (2×5 tables). Effect sizes are reported as Cramer's V for $2 \times k$ contingency tables.

For each 2×5 table, the total number of observations used in the chi-square statistic and Cramer's V is $N = 1020$ (510 manual ratings plus 510 AI ratings).

Because the study uses a within-subjects design (multiple ratings per participant and per POI), pooled contingency tests do not fully capture within-participant dependence. Statistical outcomes are therefore interpreted as descriptive evidence of distributional shifts rather than as definitive causal estimates.

3.6 Procedure

The study was conducted in three phases using a secure web platform. First, participants provided informed consent and completed a survey regarding their demographics, travel styles, and interests. This data was used to personalize the AI-generated descriptions. Second, participants evaluated 10 POIs sequentially by reading both description versions and rating them on core metrics and familiarity. This evaluation phase lasted approximately 20 to 30 minutes. Finally, participants provided overall feedback on their experience and their comfort level with AI-generated travel content. All data were collected anonymously to ensure privacy. The study was conducted in accordance with institutional ethical guidelines, with participants providing informed consent prior to participation.

3.7 Participants

A total of 51 participants were recruited through personal networks via email and direct messaging. Participants were required to be at least 18 years old and possess basic English proficiency to ensure

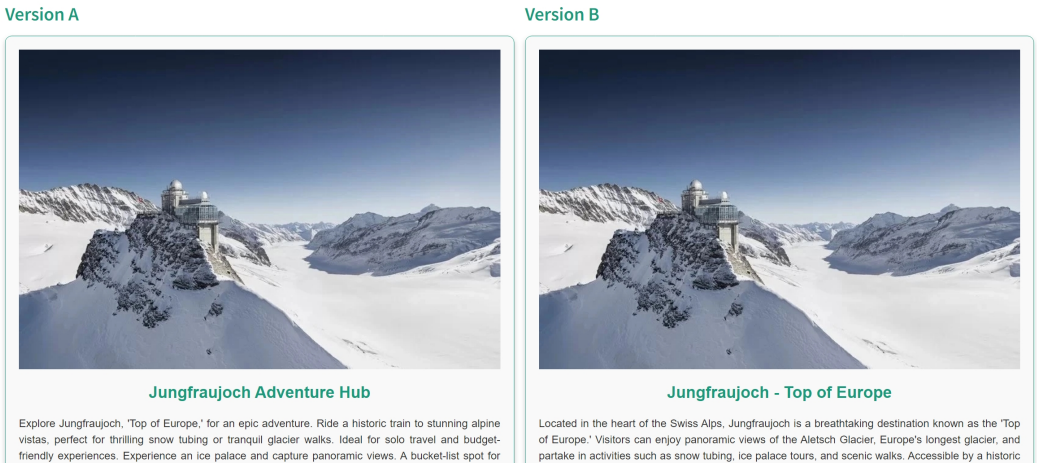

Figure 1: Evaluation interface showing two description versions (Version A and Version B) presented side by side for participant comparison

effective engagement with POI descriptions. No specific travel experience was mandated, as the study aimed to assess AI-generated descriptions across diverse user segments.

Participants ranged from 22 to 49 years old (M = 30.94, SD = 6.74), with 67% male and 33% female. The sample was predominantly Indian (n = 28) and German (n = 21), with backgrounds primarily in student and engineering roles. Most reported intermediate (n = 21) or experienced (n = 19) travel levels, providing a range of perspectives while maintaining concentration in engaged, technology-familiar demographics.

4 Results

This section presents findings from the user study evaluating LLM-adapted POI descriptions against manual baselines. Responses are summarized across three core metrics: significance, trustworthiness, and clarity. Comparative preference results and overall feedback are then reported.

Across 51 participants and 10 POIs, the study collected $N = 510$ ratings per condition for each metric (manual versus AI). Table 1 summarizes the distribution of ratings aggregated across all evaluations.

	Significance		Trust		Clarity	
Level	Man.	AI	Man.	AI	Man.	AI
1–2 (Low)	5.5	6.3	10.0	7.3	7.1	5.7
3 (Neutral)	27.1	23.1	33.3	27.4	28.8	21.4
4–5 (High)	67.4	70.6	56.7	65.3	64.1	72.9

Table 1: Distribution of User Ratings (%) for Manual vs. AI Descriptions ($N = 510$)

4.1 Perceived Significance

Participants rated both description versions on how effectively they communicated the significance and offerings of each POI using a 5-point Likert scale ranging from Strongly Disagree to Strongly Agree. Figure 2 illustrates the distribution of responses for manual and AI-generated descriptions.

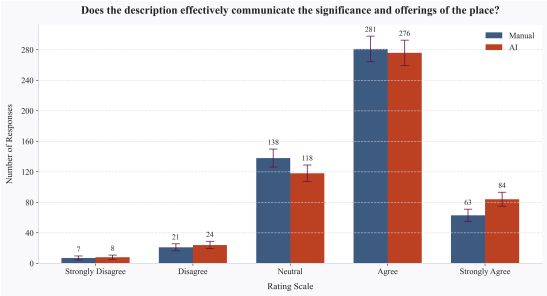

Figure 2: Comparison of perceived significance between manual and AI-generated descriptions. Error bars represent Poisson-style standard error estimates computed on raw counts ($\sqrt{n}$).

Both description types received predominantly positive ratings (Table 1). AI-generated descriptions shifted responses away from neutral toward stronger endorsement (e.g., Strongly Agree increased from 63 to 84 ratings). However, the pooled chi-square test did not indicate a statistically reliable difference in the overall distribution ($\chi^2(4) = 4.87$, $p = 0.3005$). The corresponding effect size was small (Cramer's $V = 0.069$, computed with $N = 1020$ pooled observations), suggesting similar perceived significance between conditions.

4.2 Trustworthiness

Trustworthiness was assessed on a 5-point scale ranging from Not at all to Extremely. Figure 3 presents the comparative distribution of trust ratings.

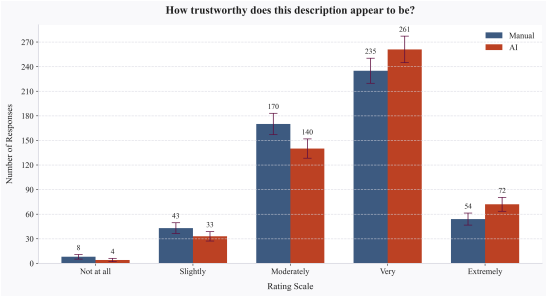

Figure 3: Comparison of perceived trustworthiness between manual and AI-generated descriptions. Error bars represent Poisson-style standard error estimates computed on raw counts ($\sqrt{n}$).

AI-generated descriptions received higher trust ratings overall, with fewer low-trust responses and more ratings in the highest category (Extremely: 72 vs. 54). The pooled chi-square test provided borderline evidence of a distributional shift ($\chi^2(4) = 9.49$, $p = 0.050$). The effect size was small (Cramer's $V = 0.096$, computed with $N = 1020$ pooled observations), indicating that the improvement is modest in magnitude and should be interpreted cautiously.

4.3 Clarity and Completeness

Participants evaluated how clear and complete the information was using a 5-point scale from Very Unclear to Very Clear. Figure 4 shows the distribution of clarity ratings.

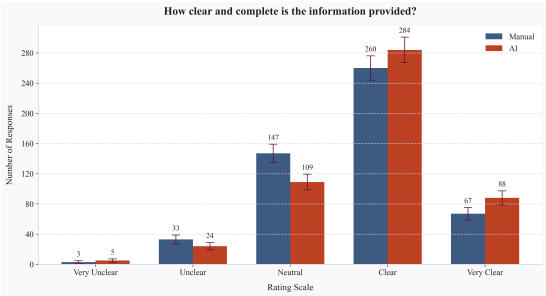

Figure 4: Comparison of perceived clarity and completeness between manual and AI-generated descriptions. Error bars represent Poisson-style standard error estimates computed on raw counts ($\sqrt{n}$).

AI-generated descriptions were rated as clearer overall, with a notable reduction in neutral responses (109 vs. 147) and more ratings in the top two categories (Clear or Very Clear: 372 vs. 327). The pooled chi-square test indicated a statistically significant difference in the distribution ($\chi^2(4) = 11.47$, $p = 0.0218$). The effect size was small (Cramer's $V = 0.106$, computed with $N = 1020$ pooled observations), consistent with a modest but systematic shift in perceived clarity.

4.4 Comparative Preferences

Beyond individual ratings, participants completed several comparative preference questions. Due to space constraints, this short paper reports the engaging comparison. Responses were nearly evenly split between AI-generated (221; 43.3%) and manual (220; 43.1%) descriptions, with 69 selections (13.5%) indicating both versions were equally engaging. This near-parity suggests that personalization increased perceived clarity without reducing engagement.

4.5 Overall Study Feedback

Following the POI evaluations, participants were invited to provide optional overall feedback. A subset of 17 participants completed this section, rating three dimensions using a 5-point scale (1 = Poor, 5 = Excellent). The overall experience with POI descriptions received a mean rating of 3.88 (SD = 0.70, Median = 4.0), indicating generally positive reception. Participants expressed moderate to strong support for the concept of automatically adapting POI descriptions based on user interests, with a mean rating of 3.75 (SD = 0.93, Median = 4.0). Comfort with reading AI-generated descriptions when planning visits to new places received the highest mean rating of 3.94 (SD = 0.83, Median = 4.0), suggesting participants were receptive to AI-generated travel content when it demonstrated quality and relevance. The relatively low standard deviations (all below 1.0) indicate consensus among participants without extreme polarization of opinions.

Across metrics, AI adaptation primarily reduced neutral ratings and increased positive categories. Given the repeated-measures structure of the data, pooled tests summarize distributional shifts.

5 Discussion

This study examined whether LLM-based personalization of POI descriptions changes how people judge the quality of travel information. Aggregated results show improved clarity ratings for AI-adapted descriptions, with borderline evidence of improved trustworthiness ($p = 0.050$). Across measures, AI adaptation primarily reduced neutral responses and increased positive categories. This pattern is consistent with increased perceived relevance: when descriptions foreground details aligned with a reader's interests, the same factual content can appear easier to process and more useful.

Effect sizes were small (Cramer's V in the 0.07 to 0.11 range), suggesting modest practical gains. In addition, statistical tests were computed on pooled ratings, while the study design includes repeated judgments by the same participant across POIs. Outcomes should therefore be interpreted as descriptive evidence of distributional shifts rather than definitive estimates. When considering multiple outcomes (significance, trustworthiness, clarity), uncorrected p-values should be treated cautiously.

From a practical perspective, these findings support the use of LLMs to adjust emphasis and framing in POI micro-content using lightweight preference signals. This approach can scale personalization without extensive manual editing. Platforms should treat

transparency and disclosure as design choices, setting clear expectations without unnecessarily signaling low quality.

5.1 Limitations and Future Work

Several limitations should be acknowledged. Participants evaluated descriptions in a controlled setting rather than during real-world trip planning, limiting ecological validity. The study measured perceived outcomes but not downstream behavior such as booking rates or actual visits. The within-subjects design may introduce demand characteristics, and the sample composition limits generalizability to other demographics and cultural contexts. Future work should expand participant diversity, conduct longitudinal field studies tracking actual behavior, and test finer personalization factors (trip purpose, companions, budget). Investigating how transparency about AI involvement affects trust, and balancing personalized relevance with serendipitous discovery, remain open questions.

6 Conclusion

This study provides evidence that large language models can personalize point-of-interest descriptions in ways that improve how users judge travel information. Compared to manual baselines, AI-adapted descriptions were rated as clearer and showed borderline evidence of improved trustworthiness, while perceived significance remained comparable. Optional post-study feedback suggests that participants were generally comfortable reading AI-generated descriptions in this setting.

These findings suggest that personalization can support travel decision-making by foregrounding details that match user interests while preserving core facts about a place. For travel platforms, LLM-based rewriting offers a practical path to scaling tailored micro-content without extensive manual editing. More broadly, results highlight that description framing can shape perceived information quality alongside which places are recommended.

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